

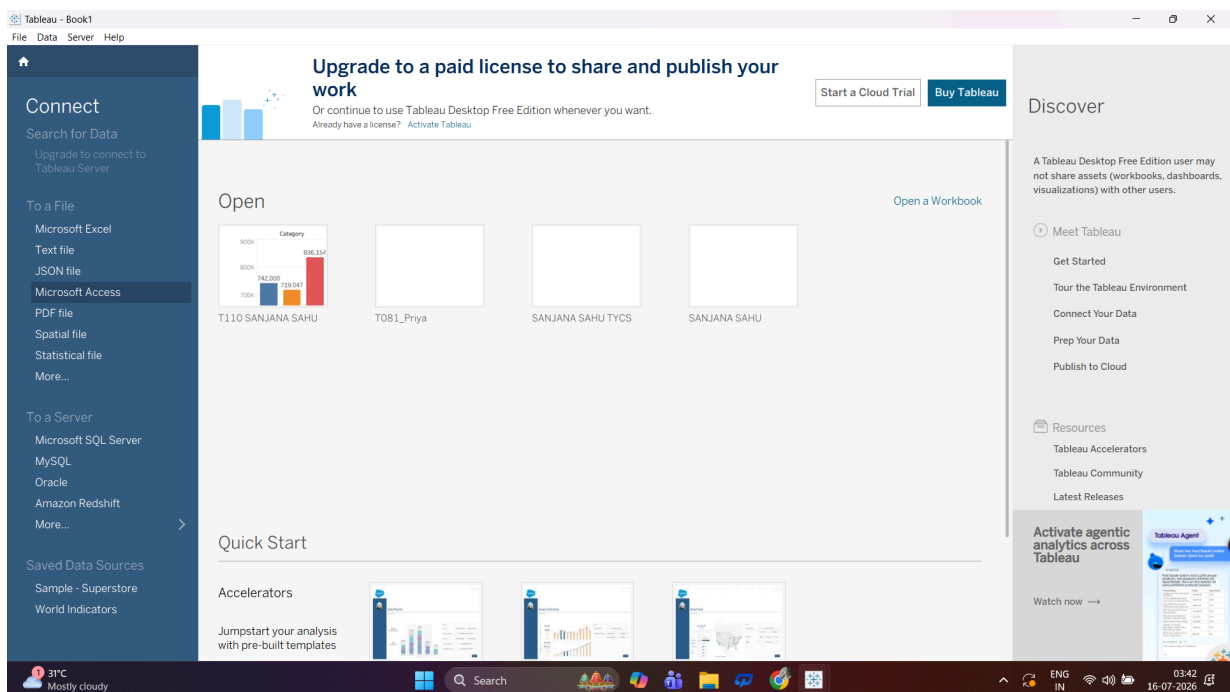
Practical 2: Working with Data Sources and Data Preparation in Tableau

- A. Connect to different data sources (Excel, CSV, Database)
- B. Create and use data extracts vs live connections
- C. Manage metadata (rename, hide, change data types)
- D. Perform joins (inner, left, right) on multiple tables
- E. Apply data blending between two sources
- F. Implement different types of filters (dimension, measure, date)
- G. Optimize performance using extracts and filtering

Dataset: E-commerce Dataset

Step A: Connecting to Different Data Sources

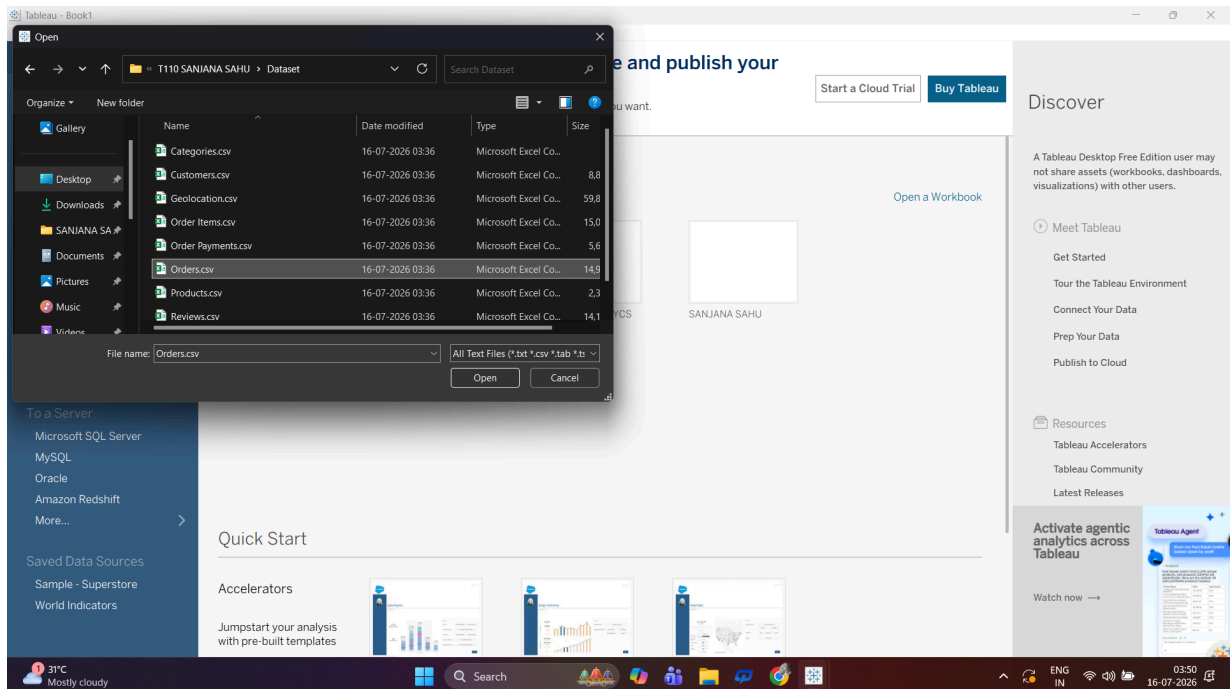
1. Launch the Tableau Desktop application.



2. Under the To a File sub-header, click on Text File (Tableau classifies .csv spreadsheet extensions as flat text files)

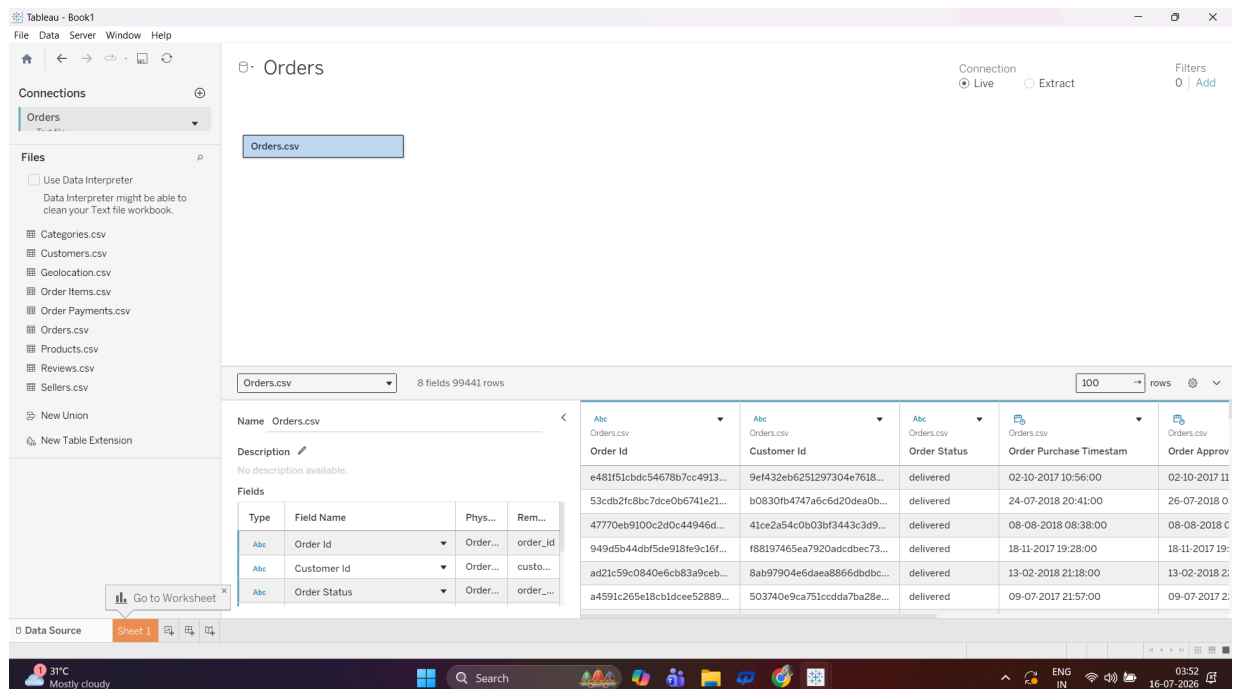
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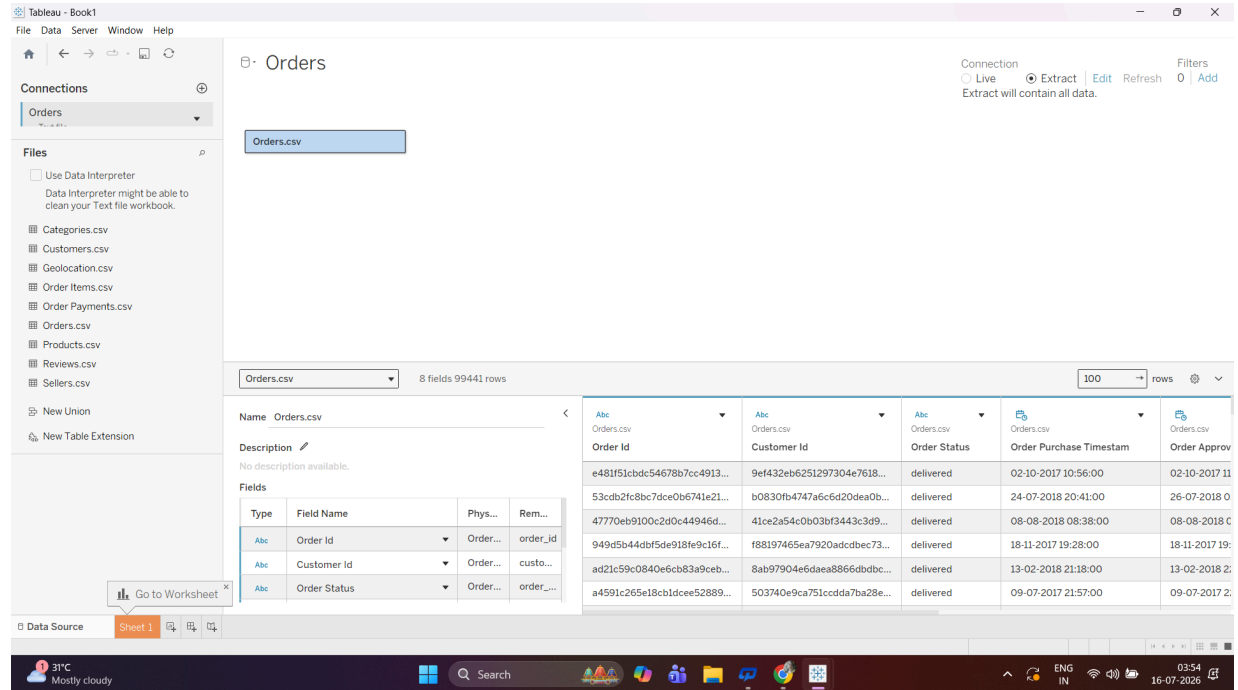
Step B: Creating Data Extracts vs. Live Connections

1. Once the dataset populates the main staging view, navigate to the top-right corner of the Data Source interface.
2. Locate the connection mode toggle section.

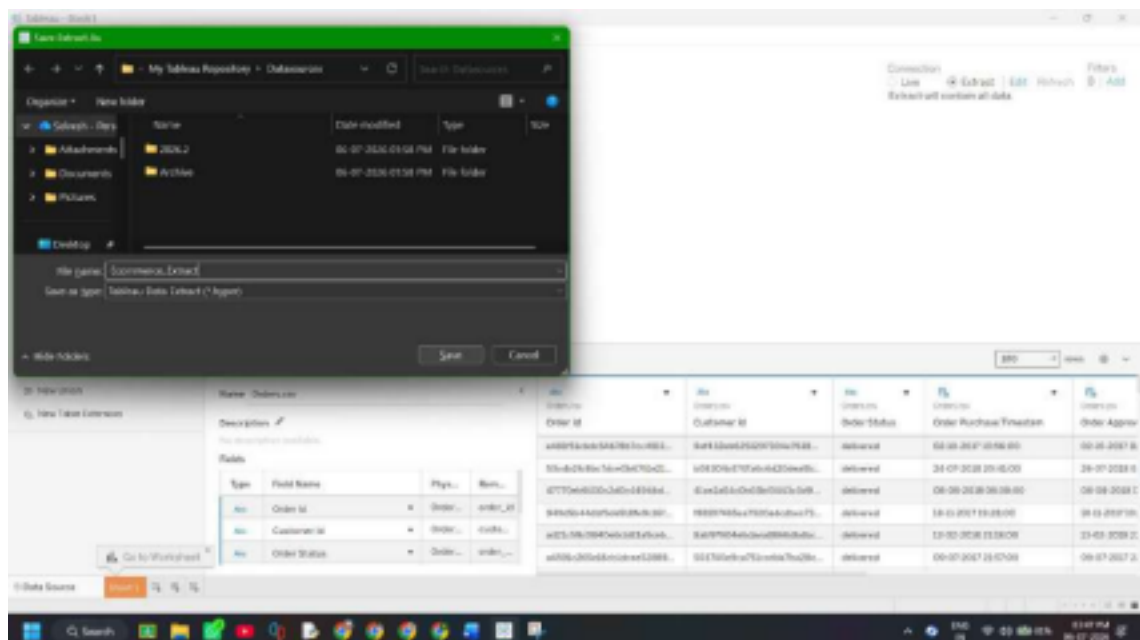


3. Switch the active selection from Live over to Extract.

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4. Click on the Sheet 1 navigation tab located at the bottom-left corner of the workspace screen.
5. A system dialog save window will immediately appear. Name the local snapshot file Ecommerce_Extract and click Save.

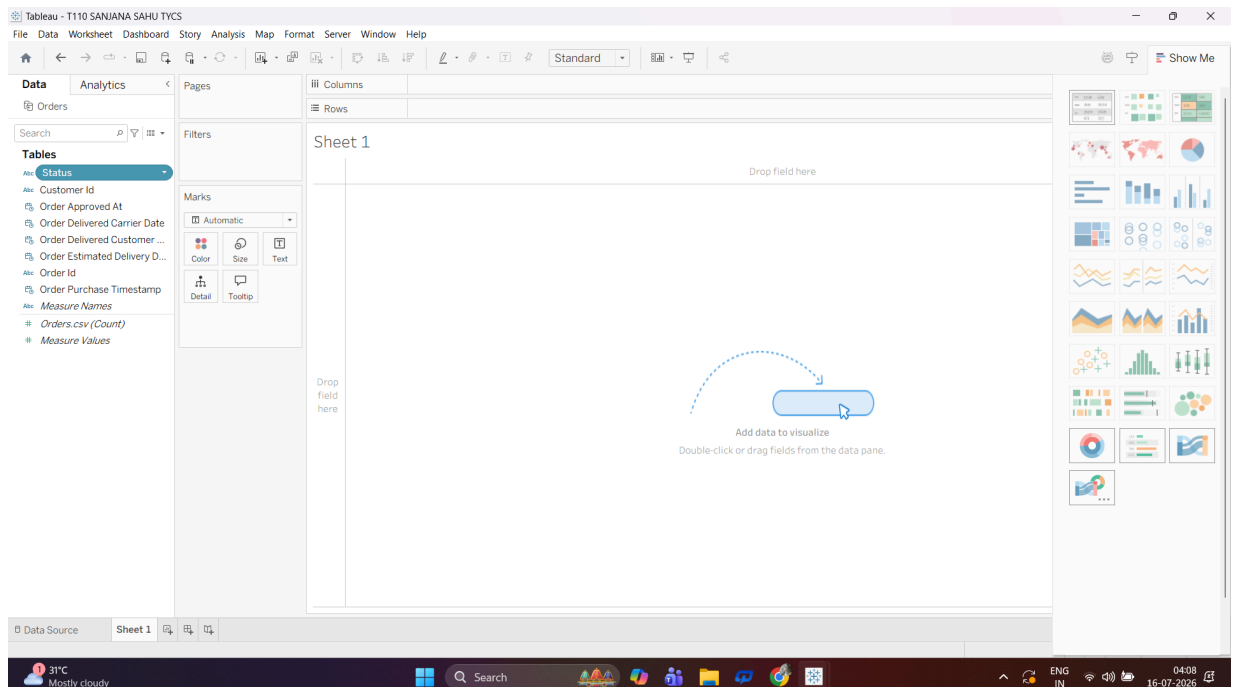
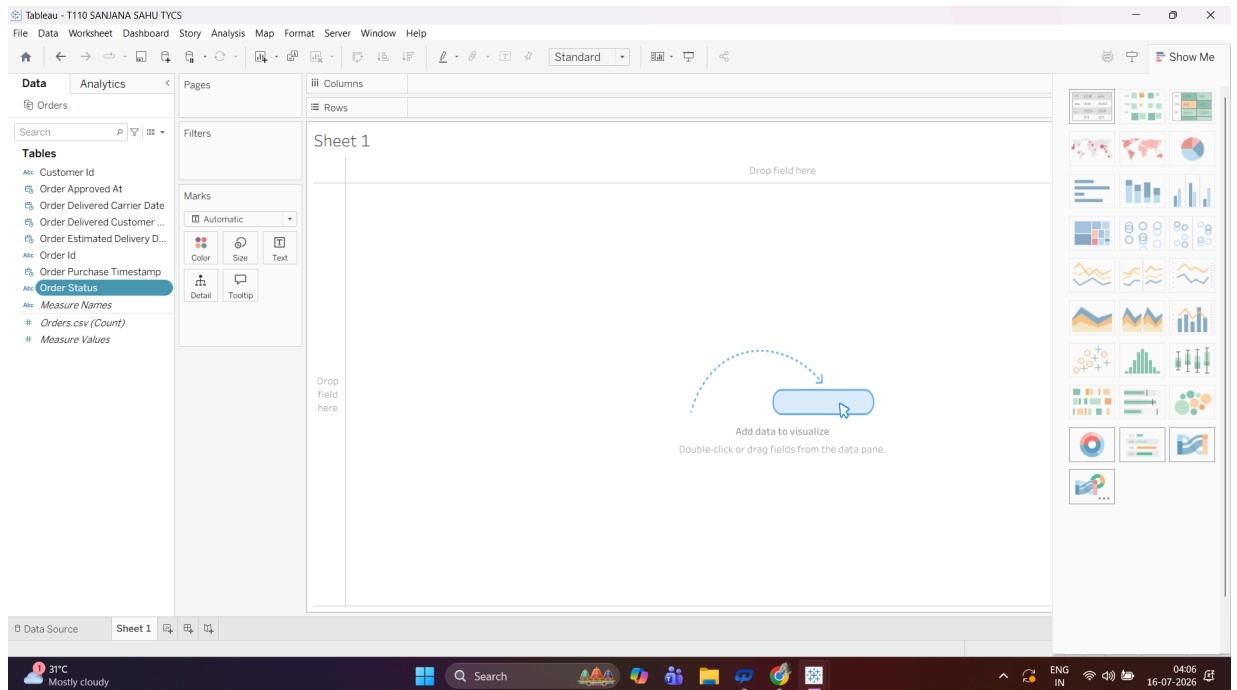


Step C: Managing Operational Metadata

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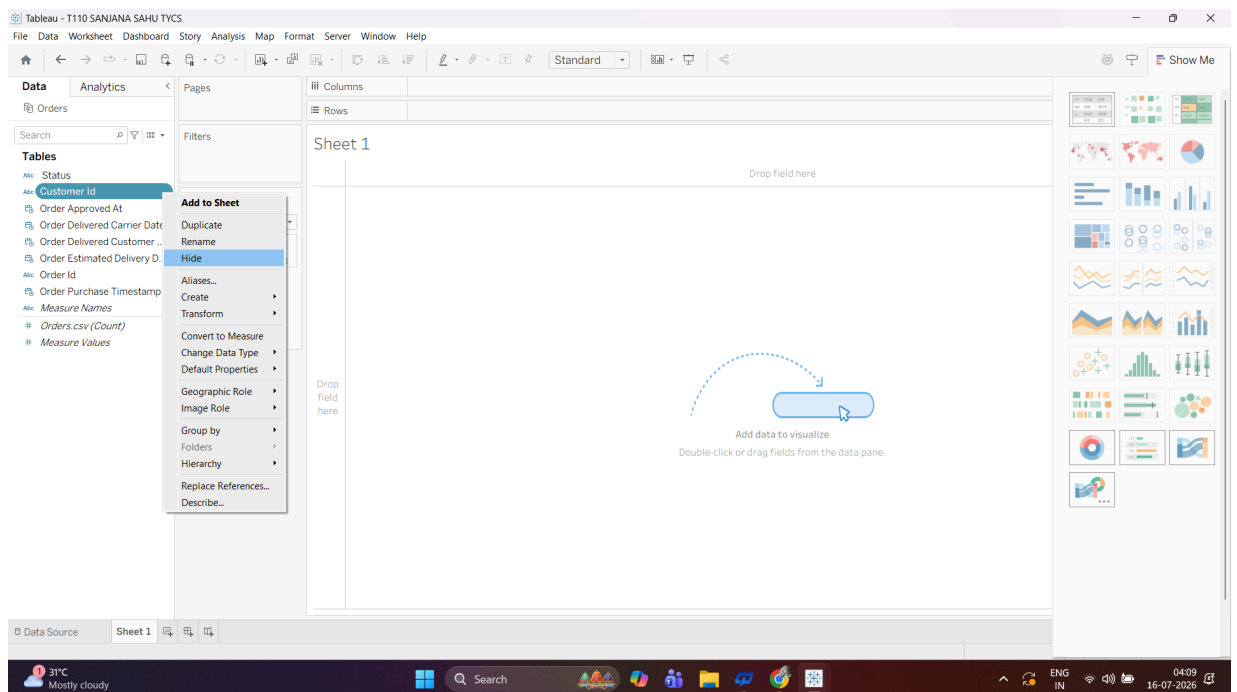
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1. Navigate to the blank workspace canvas on Sheet 1 and view the left-hand side Data Pane.
2. Rename a Field: Locate the dimensional attribute field labeled order_status or Order Status. Right-click the field name, select Rename, type in Status, and hit Enter.



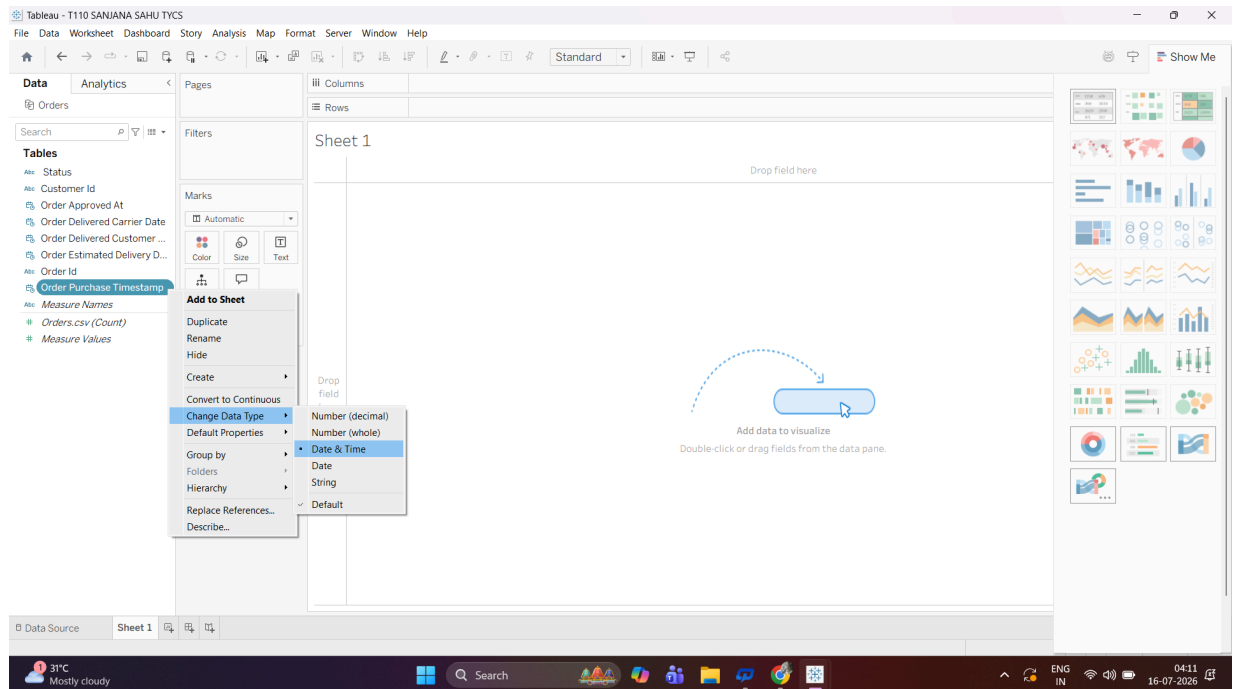
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3. Hide a Field: Locate an internal tracking string variable that will not be used in visual reporting, such as customer. Right-click the field name and select Hide.



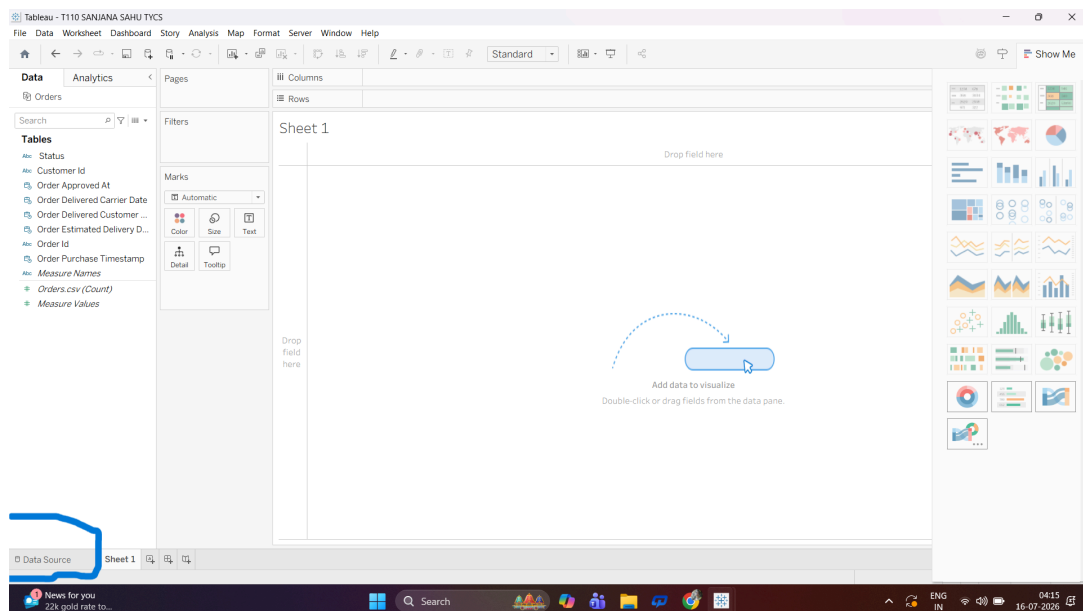
4. Change a Data Type: Locate the operational timestamp column labeled order_purchase_timestamp. Click directly on the data type icon (which may incorrectly display as text Abc or a calendar date) positioned immediately to the left of the field name. Select Date & Time from the context dropdown menu to force proper structural recognition (Note: It may be by default selected to Date and Time).

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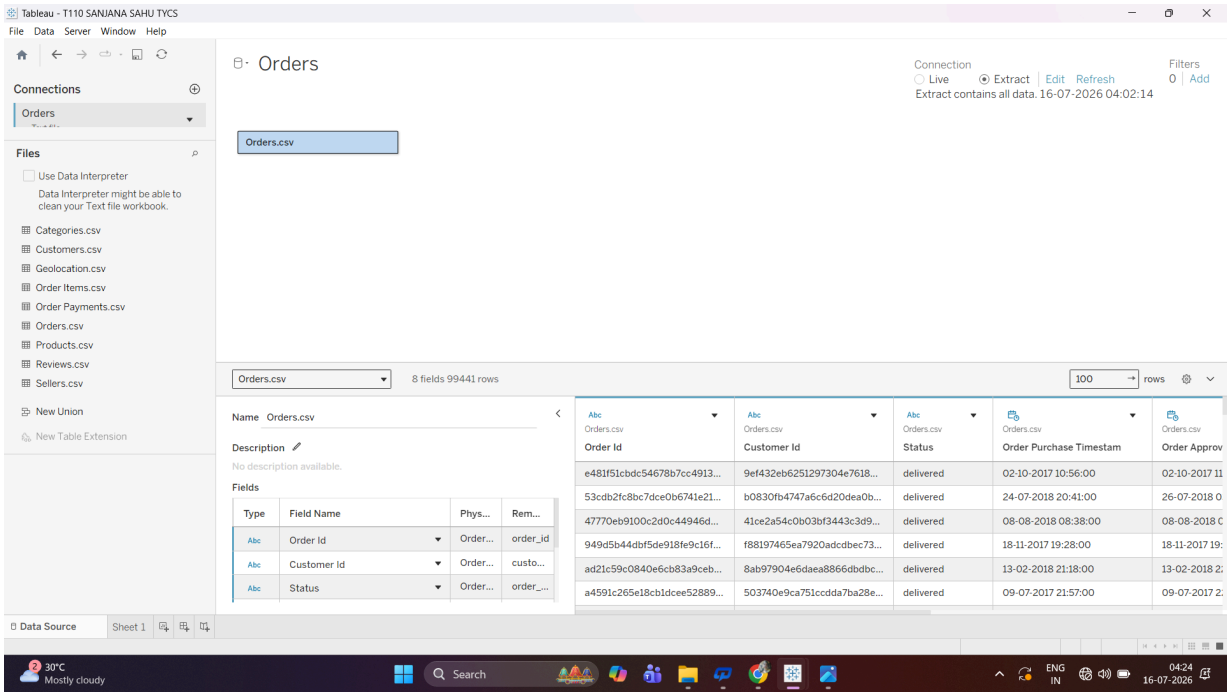
Step D: Performing Relational Joins on Multiple Tables

1. Click the Data Source tab positioned at the bottom-left corner of the window to return to the data schema designer.



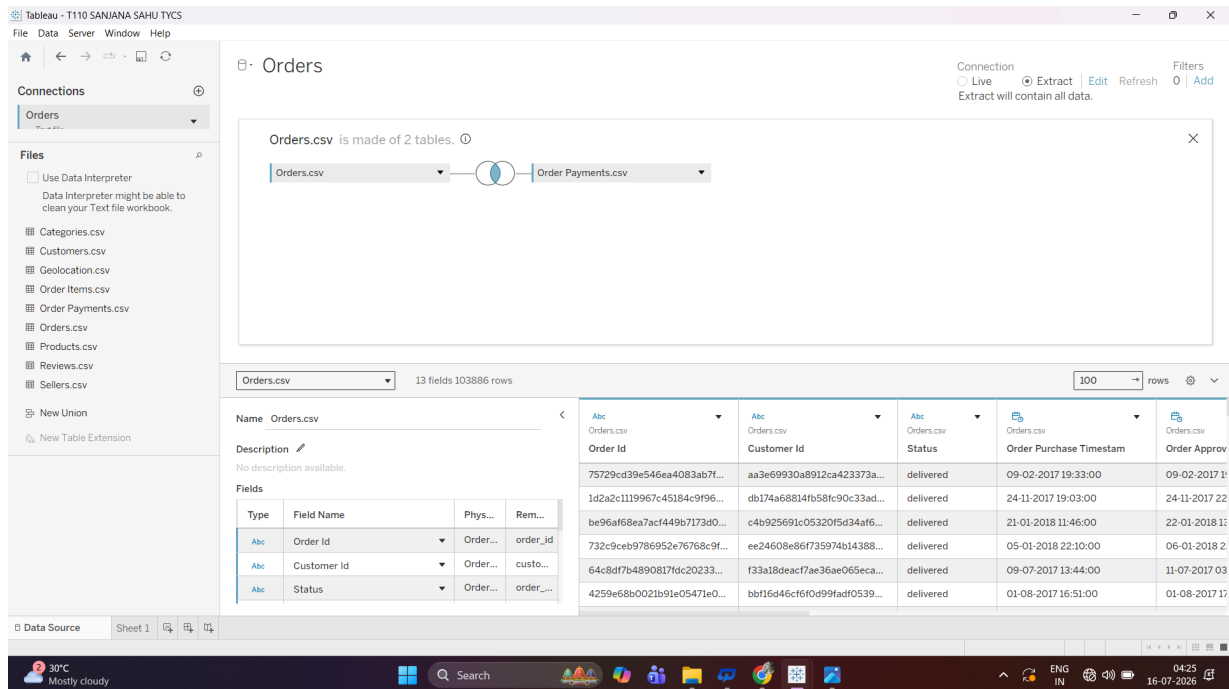
2. In the top staging canvas area, locate the rectangular Orders.csv database block.

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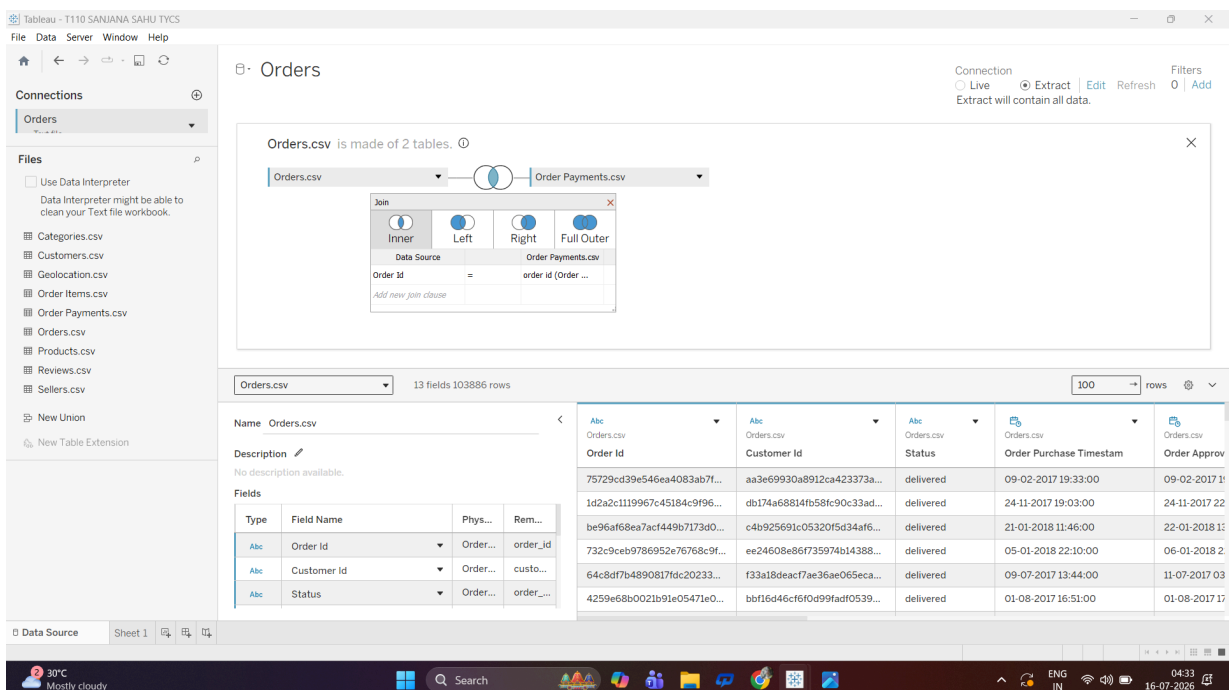
3. Double-click directly on the Orders.csv block. This action drills down through the logical relationship layer and enters the Physical Layer canvas space.

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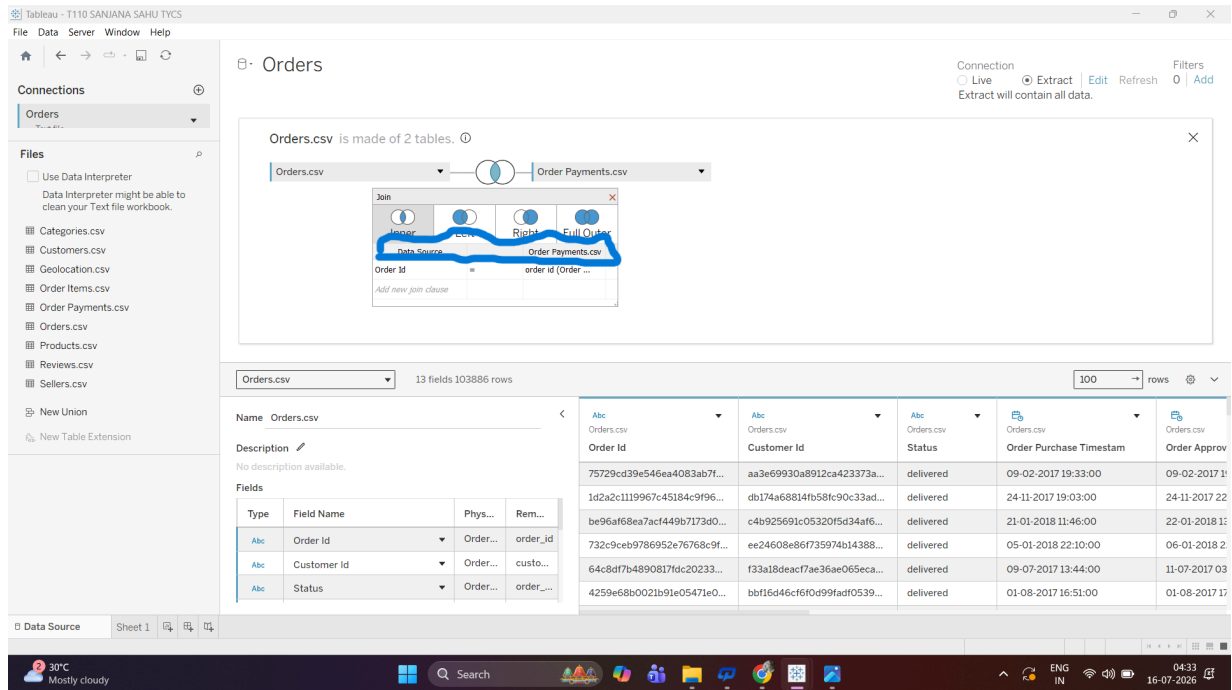
4. From the left-hand files listing pane, click and drag the Order Payments.csv file, dropping it directly next to the active Orders.csv block. A traditional Venn diagram icon link will snap open between them.

5. Click directly on the Venn diagram icon to open the Join configuration properties box.



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6. Under the data grid coupling properties, select `order_id` as the matching data key column drop-down for both the left and right tables (Note: in some case Tableau automatically selects the Order Id).



The screenshot shows the Tableau interface with a join configuration between `Orders.csv` and `Order Payments.csv`. The join is set to 'Inner' and the matching field is 'order_id'. A data preview window at the bottom shows the resulting data grid.

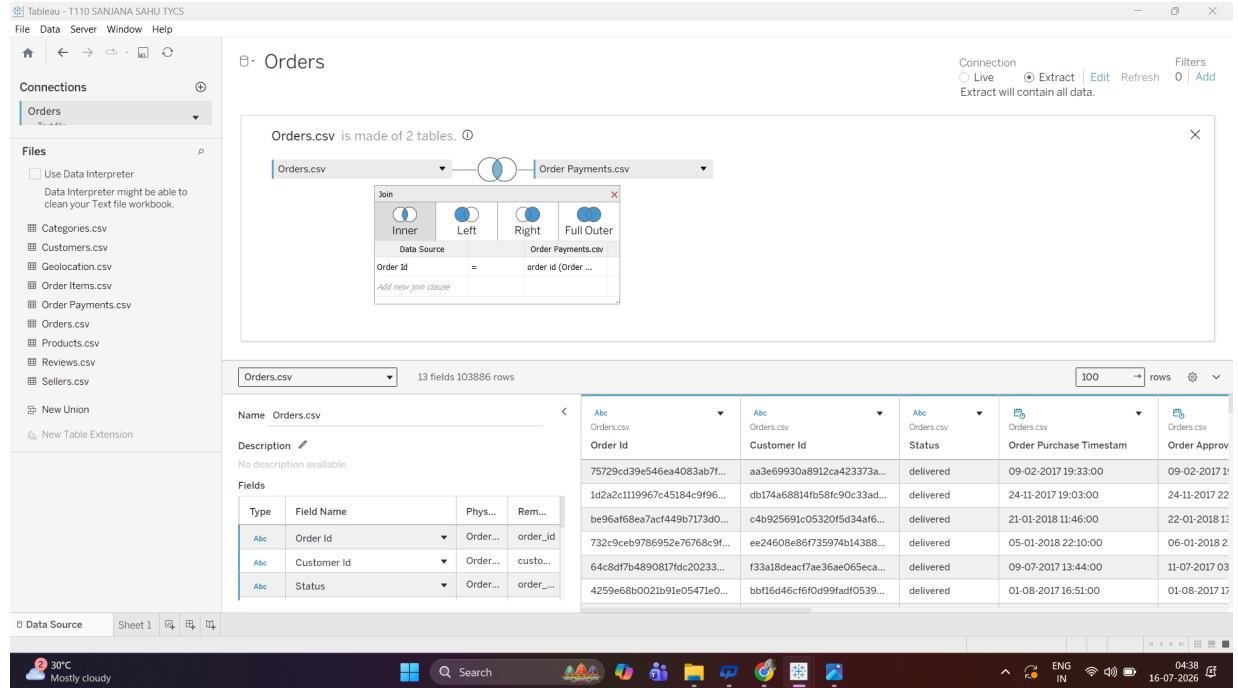
Type	Field Name	Phys...	Rem...
ABC	Order Id	Order...	order_id
ABC	Customer Id	Order...	custo...
ABC	Status	Order...	order_...

Order Id	Customer Id	Status	Order Purchase Timestamp	Order Approv
75729cd39e546ea4083ab7f1...	aa3e69930a8912ca423373a...	delivered	09-02-2017 19:33:00	09-02-2017 11:00:00
1d2a2c1119967c45184c9f96...	db174a68814fb58fc90c33ad...	delivered	24-11-2017 19:03:00	24-11-2017 22:00:00
be96af68ea7ac449b7173d0...	c4b925691c05320f5d34af6...	delivered	21-01-2018 11:46:00	22-01-2018 11:00:00
732c9ceb9786952e76768c9f...	ee24608e86f735974b14388...	delivered	05-01-2018 22:10:00	06-01-2018 22:00:00
64c8df7b4890817f6c20233...	f33a18deacf7ae36ae065eca...	delivered	09-07-2017 13:44:00	11-07-2017 03:00:00
4259ee68b0021b91e05471e0...	bbf16d46cf6f0d99fadf0539...	delivered	01-08-2017 16:51:00	01-08-2017 17:00:00

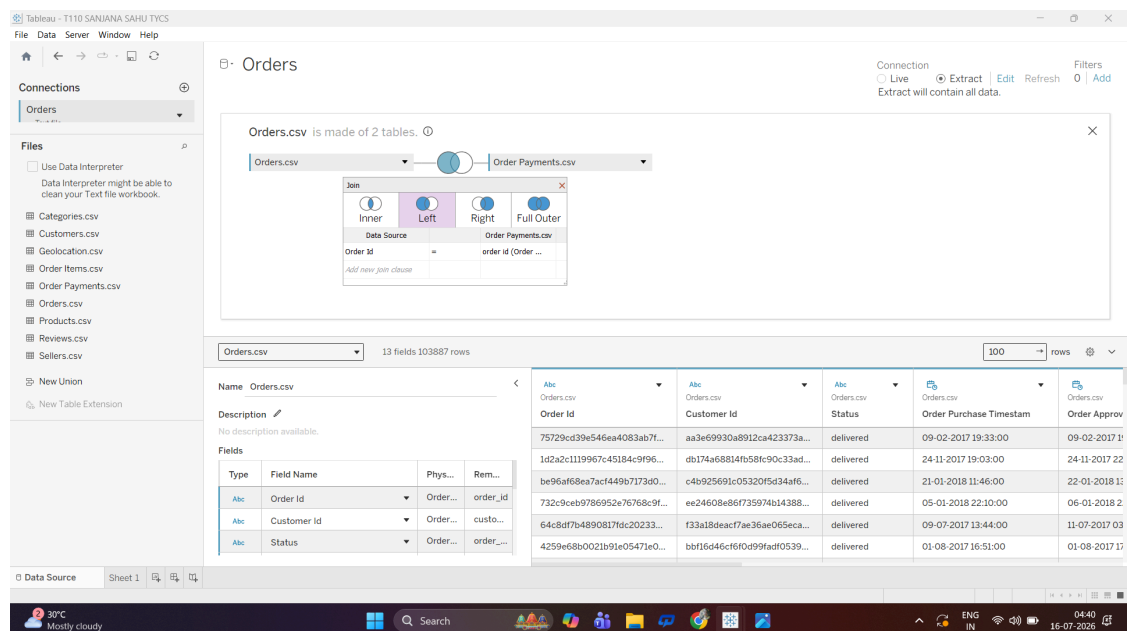
7. Toggle Join Configurations:

- o Click Inner: Observe the underlying data preview window at the bottom. The rows instantly filter to show only records that contain both an order entry and a corresponding transaction payment receipt.

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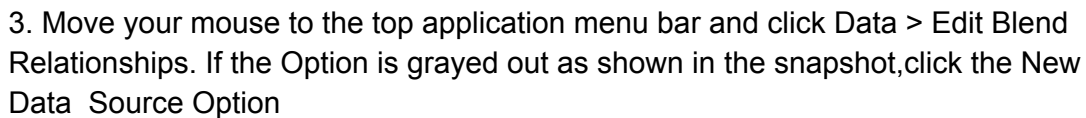


- o Click Left: Notice the view updates to safely preserve 100% of the root rows coming from Orders.csv, leaving payment parameter cells completely empty if an order has no matching payment record.

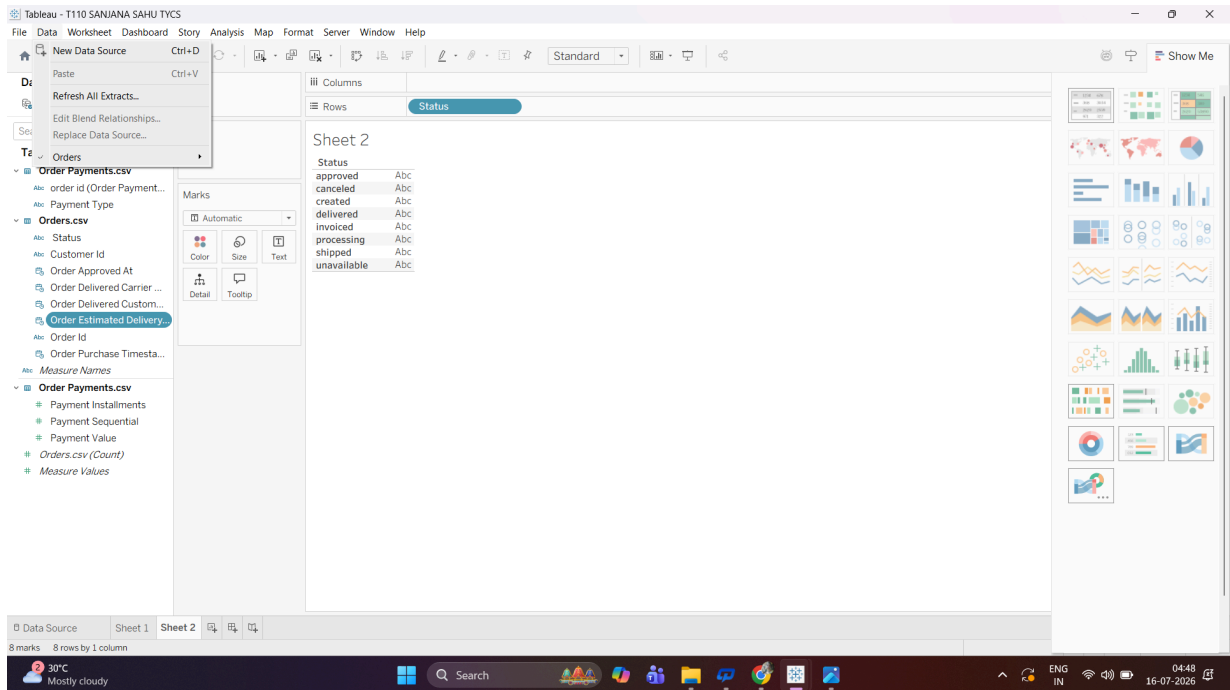


Step E: Applying Data Blending Between Independent Sources

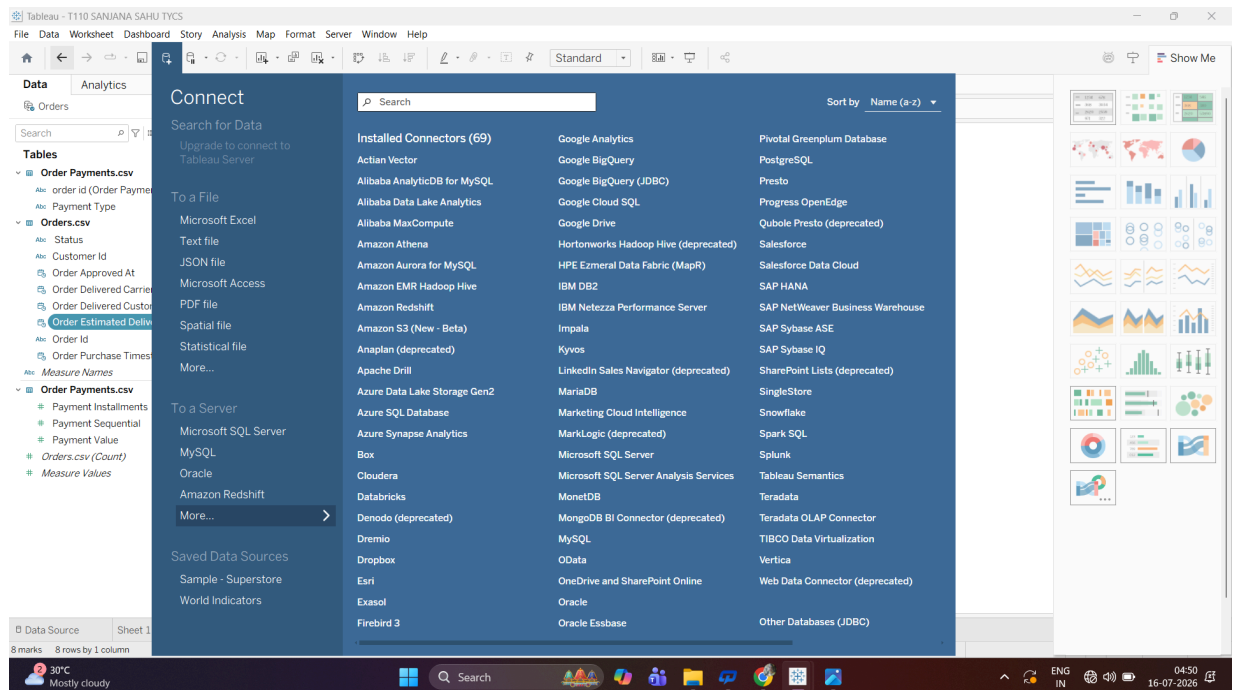
1. Create a brand new blank workspace sheet by clicking on the New Worksheet icon (the chart block containing a \$\$ symbol) positioned at the bottom navigation tab area. This opens Sheet 2.
2. Ensure your joined Orders dataset connection is highlighted at the top left of the Data Pane. Drag the newly modified Status dimension field and drop it onto the Rows shelf. A blue checkmark will appear on the source, designating it as the Primary data pipeline.



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After Clicking new data source you are directed to blue menu window select text file to add the Customers.csv file.



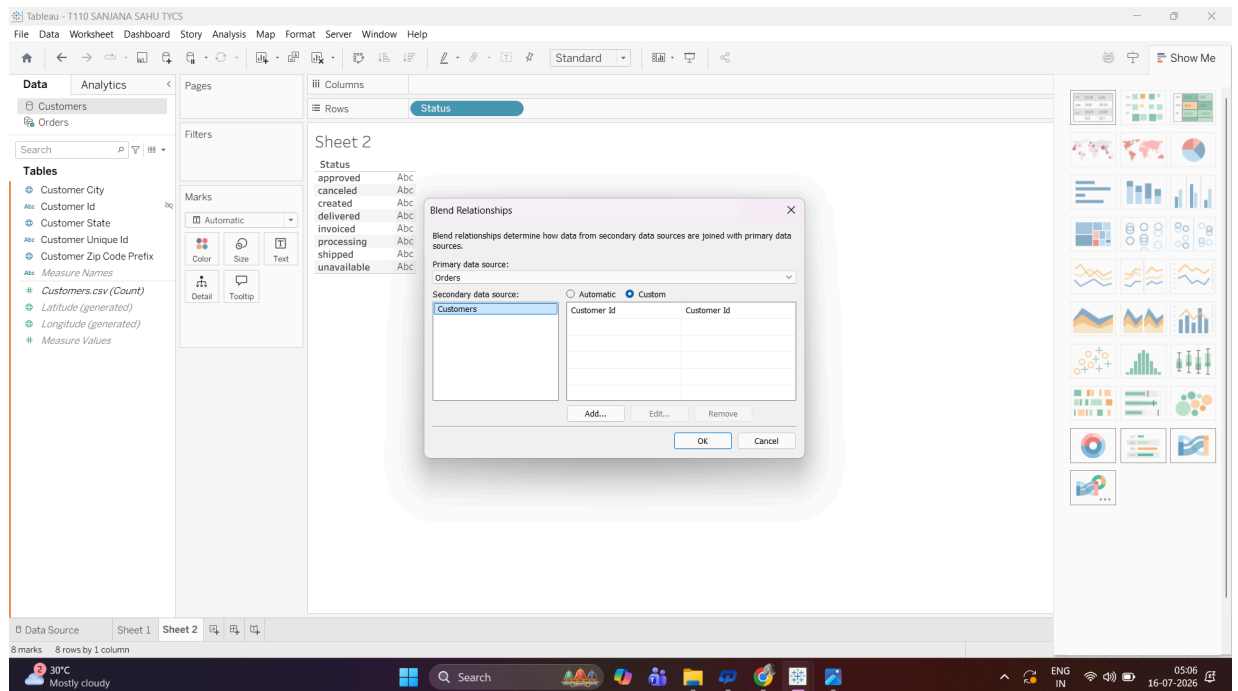
After adding the CSV file get back to sheet 2 and recheck the Edit Blend Relationship Option and click on it

The screenshot shows the Tableau Desktop application window. The top menu bar includes File, Data, Worksheet, Dashboard, Story, Analysis, Map, Format, Server, Window, and Help. The 'Data' menu is open, displaying options like 'New Data Source', 'Paste', 'Refresh All Extracts...', 'Edit Blend Relationships...', and 'Replace Data Source...'. The 'Edit Blend Relationships...' option is highlighted. The main workspace shows 'Sheet 2' with a table of data. The table has two columns: 'Status' and 'ABC'. The 'Status' column contains values: approved, canceled, created, delivered, invoiced, processing, shipped, and unavailable. The 'ABC' column contains the value 'ABC' for all rows. The bottom status bar indicates '8 marks' and '8 rows by 1 column'.

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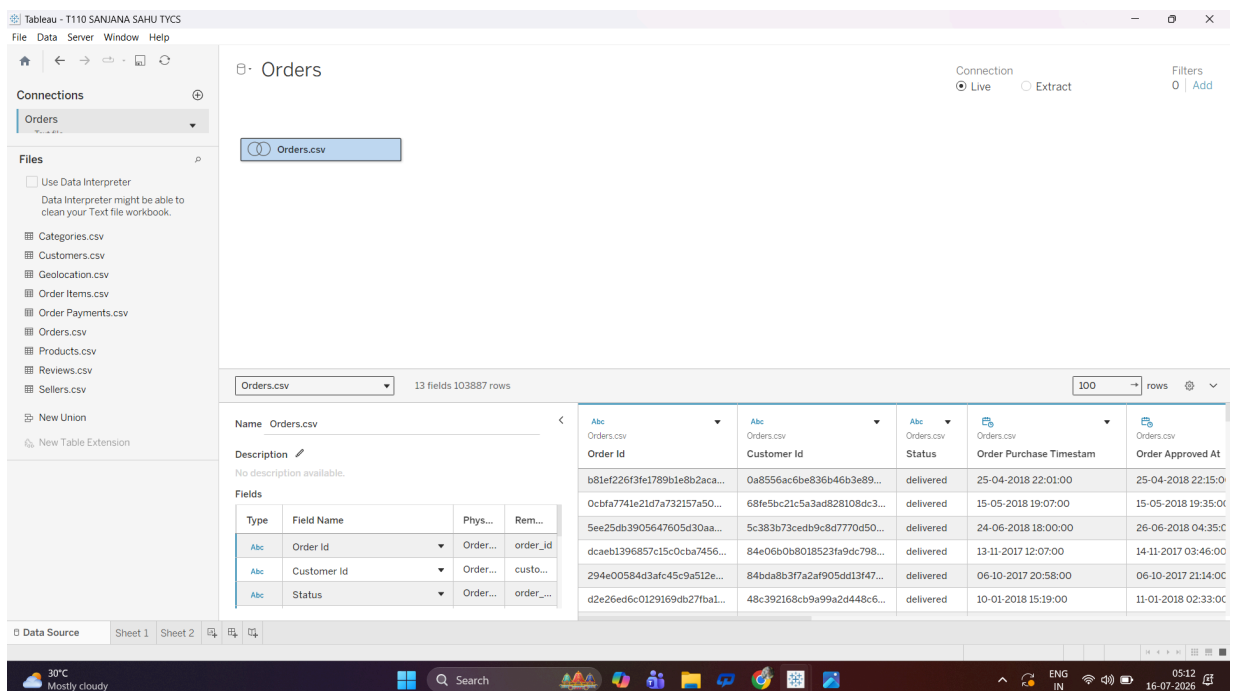
Practical 2: Working with Data Sources and Data Preparation in Tableau



If the Customer Id is not visible in primary data its not an issue since we hid it earlier at

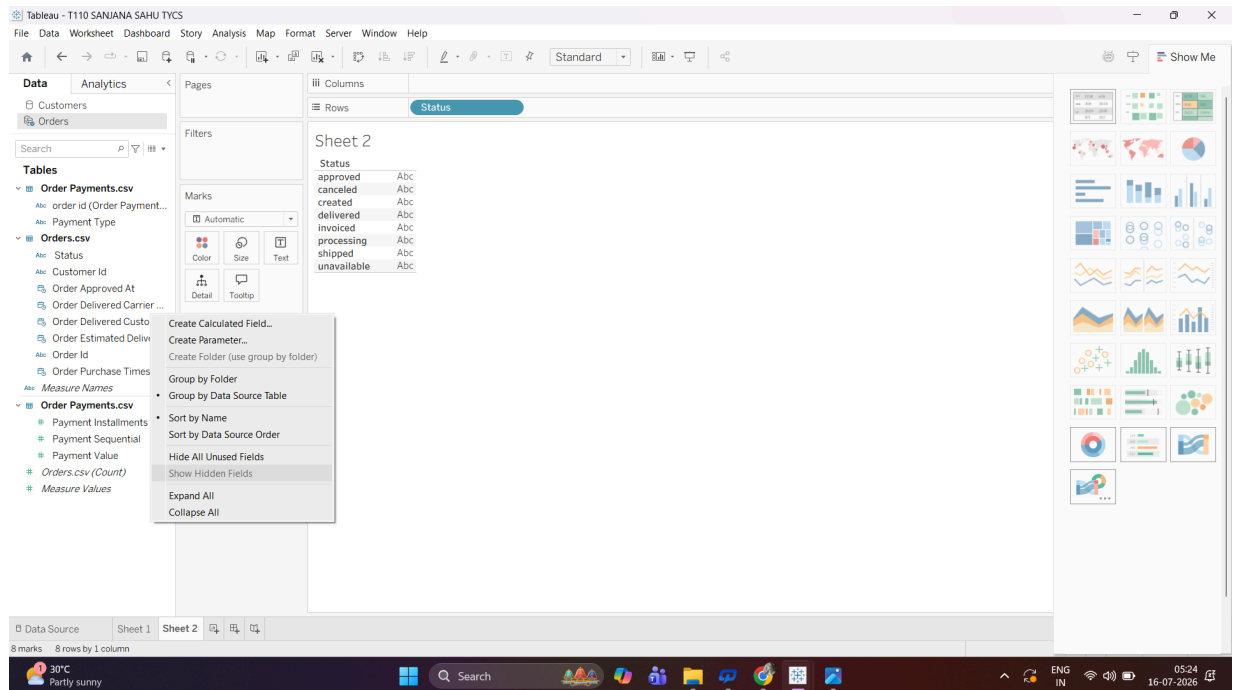
Step C

Now Navigate Back to data source and change the connection to “LIVE” from “EXTRACT”

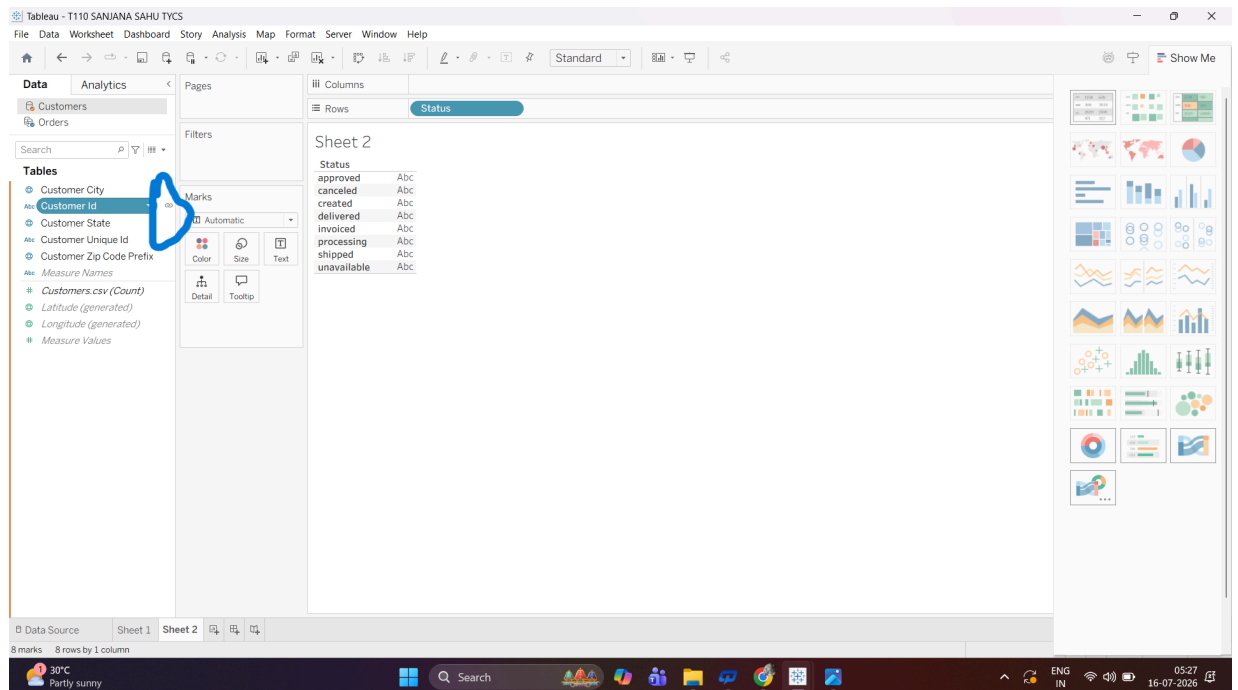


Now Go Back to any to the Sheet 2 and right click on the Data Pane's empty space and click on “Show Hidden Fields”

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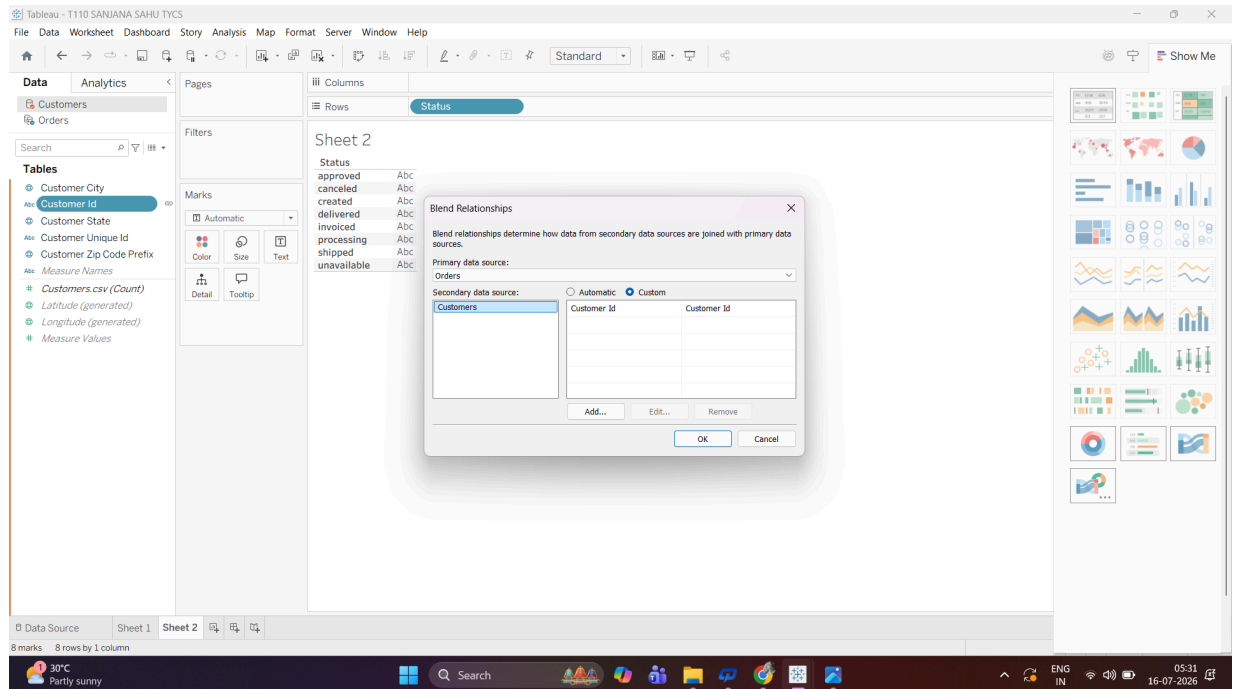


After Clicking it the Customer Id comes back into the data pane and make sure you click the “eye button” next to it marked in bellow provided snapshot

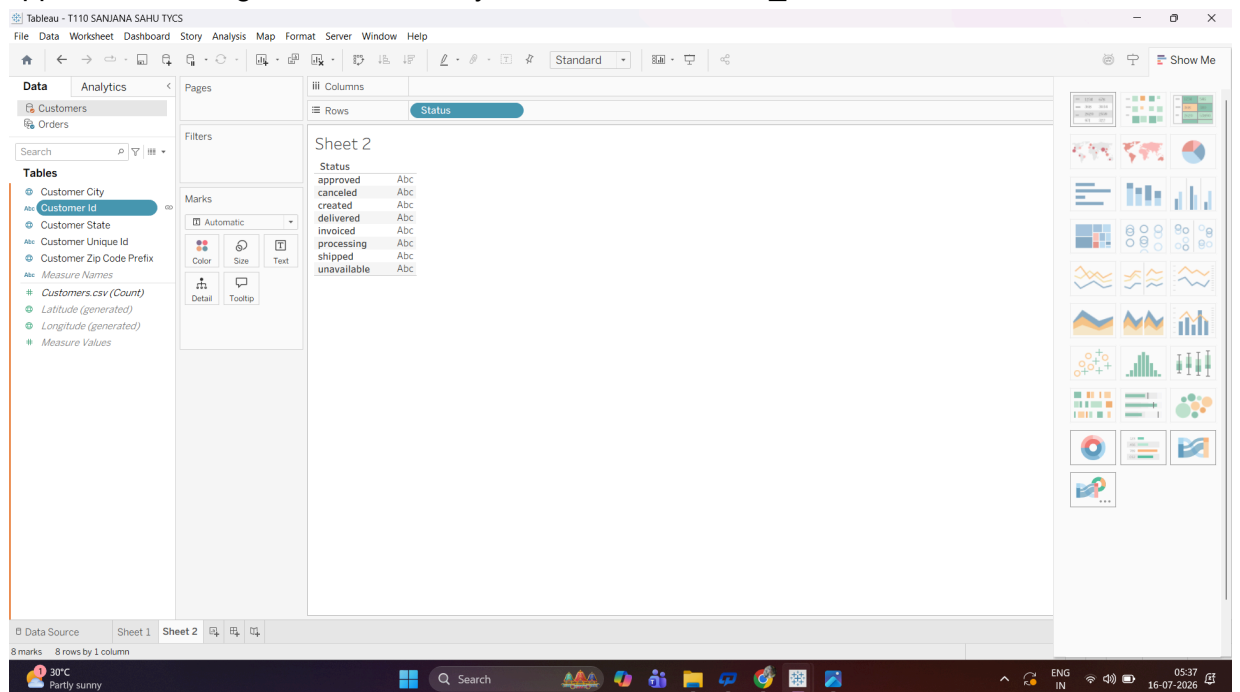


Now go back to Edit Blend Relationship option and after clicking it you may see both Customer Id in Left and Right, After that click on Custom it changes from automatic to custom and click Ok

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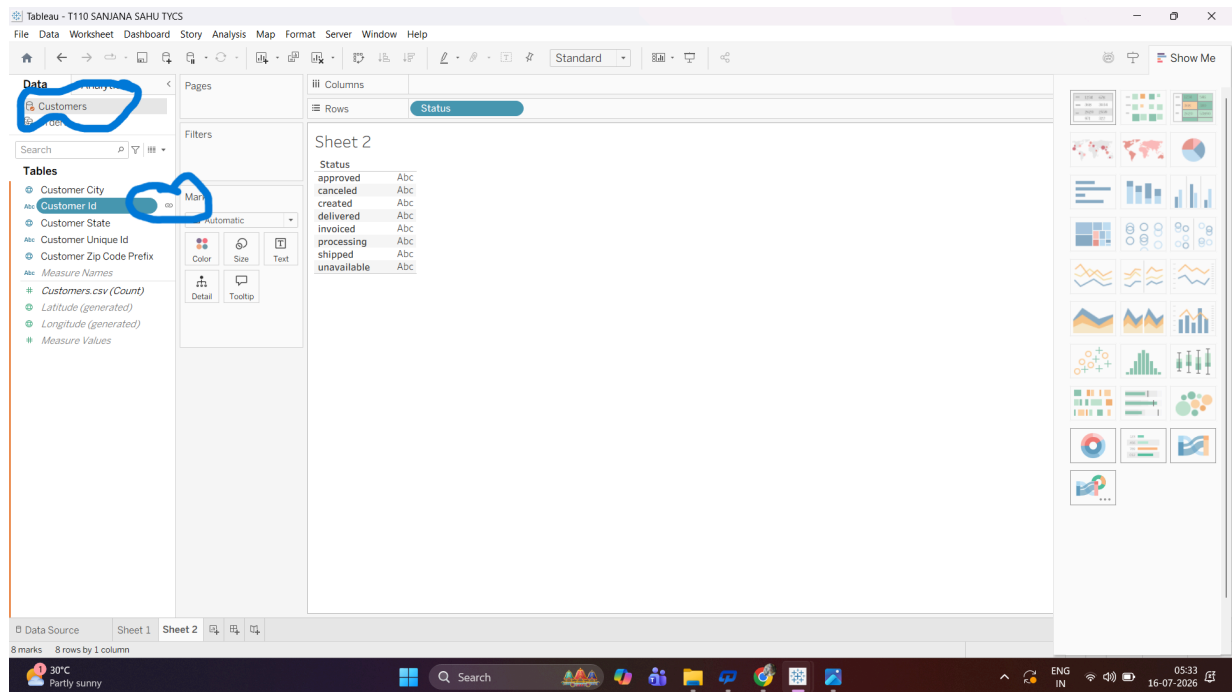


6. Now, go back to the top-left section of your Data Pane and click directly on the separate Customers dataset tracking label. An orange chain link icon will now appear locked together immediately next to the customer_id field

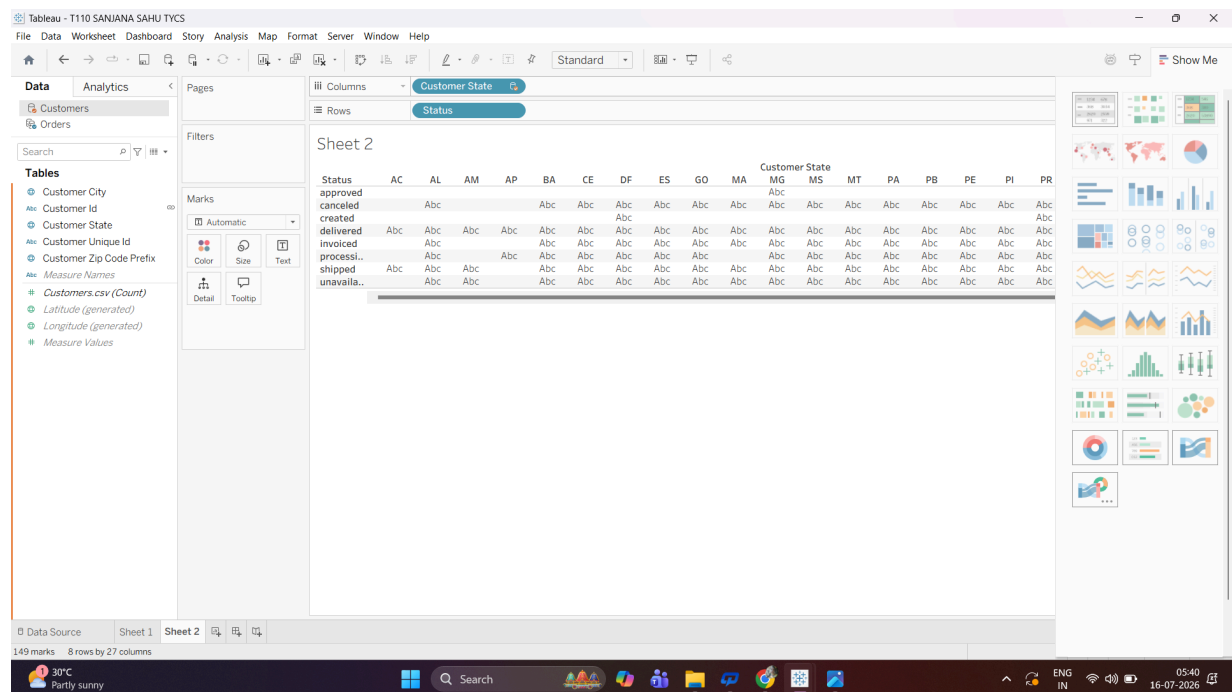


At the start you may not see the orange tick on the Customer database for that click on the unlinked chain next to the customer id on the data pane after clicking the orange tick appear as shown in the snapshot

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7. Click and drag the customer_state dimension variable out from this secondary dataset list and drop it onto the Columns shelf.

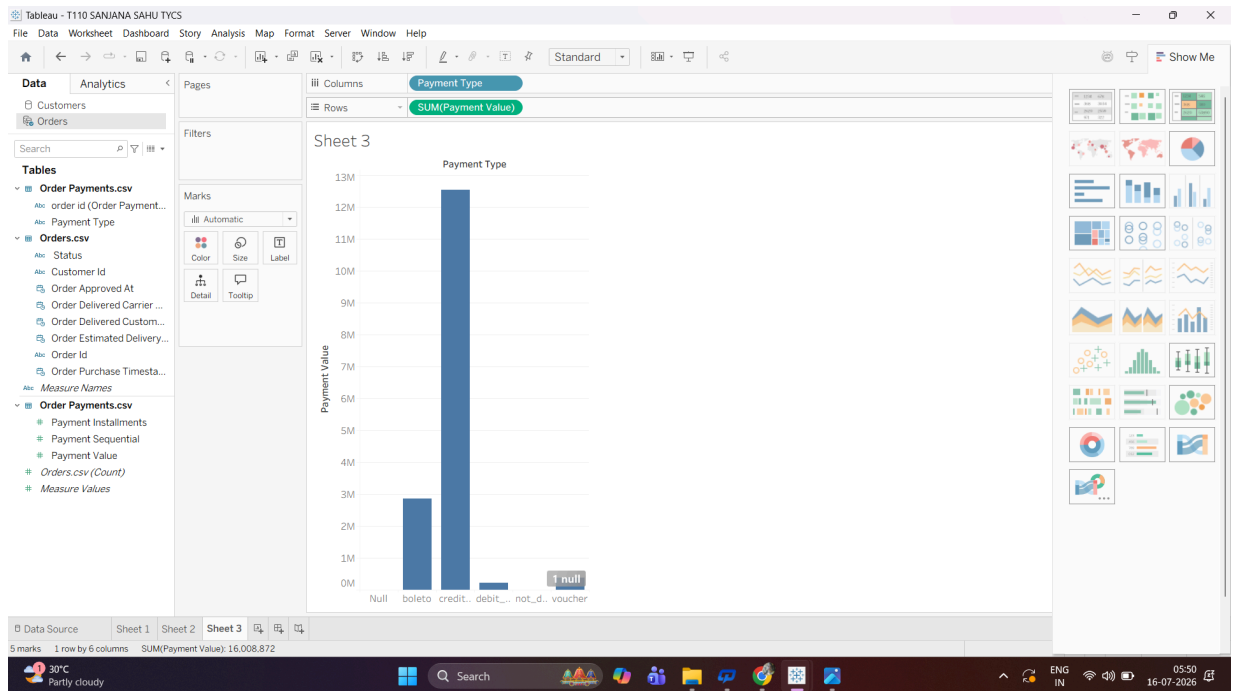


Step F: Implementing Multi-Tiered Filters

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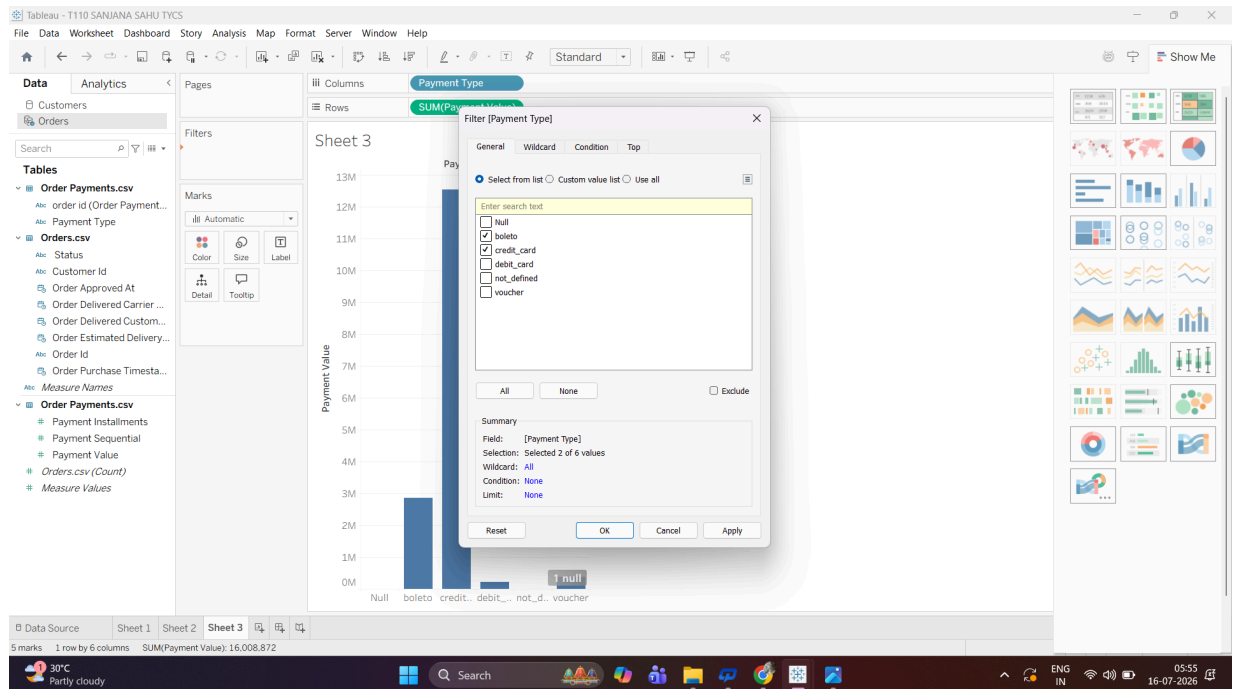
Practical 2: Working with Data Sources and Data Preparation in Tableau

1. Open a clean worksheet. Drag the quantitative measure field `payment_value` onto the Rows shelf and place the categorical dimension `payment_type` onto the Columns shelf.

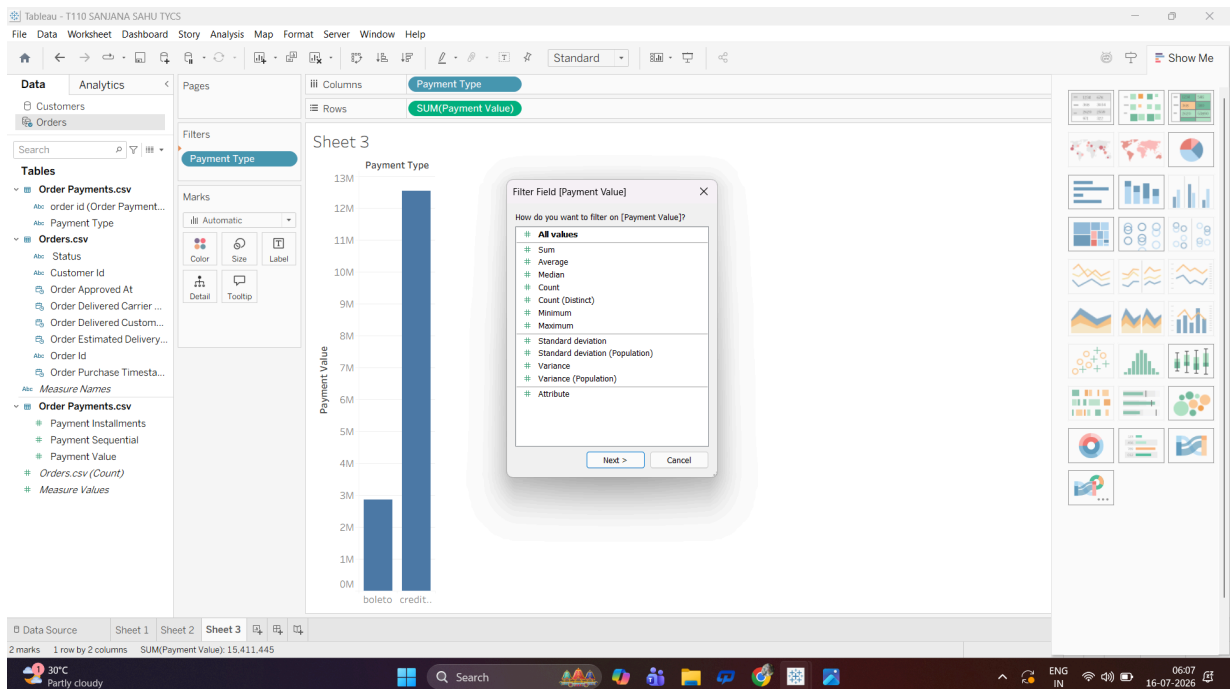


2. Dimension Filter: From your Data Pane list, drag the categorical string attribute `payment_type` and drop it onto the empty Filters shelf positioned immediately above the Marks card layout. Uncheck the box for `not_defined`, check the boxes for `credit_card` and `boleto`, and click OK.

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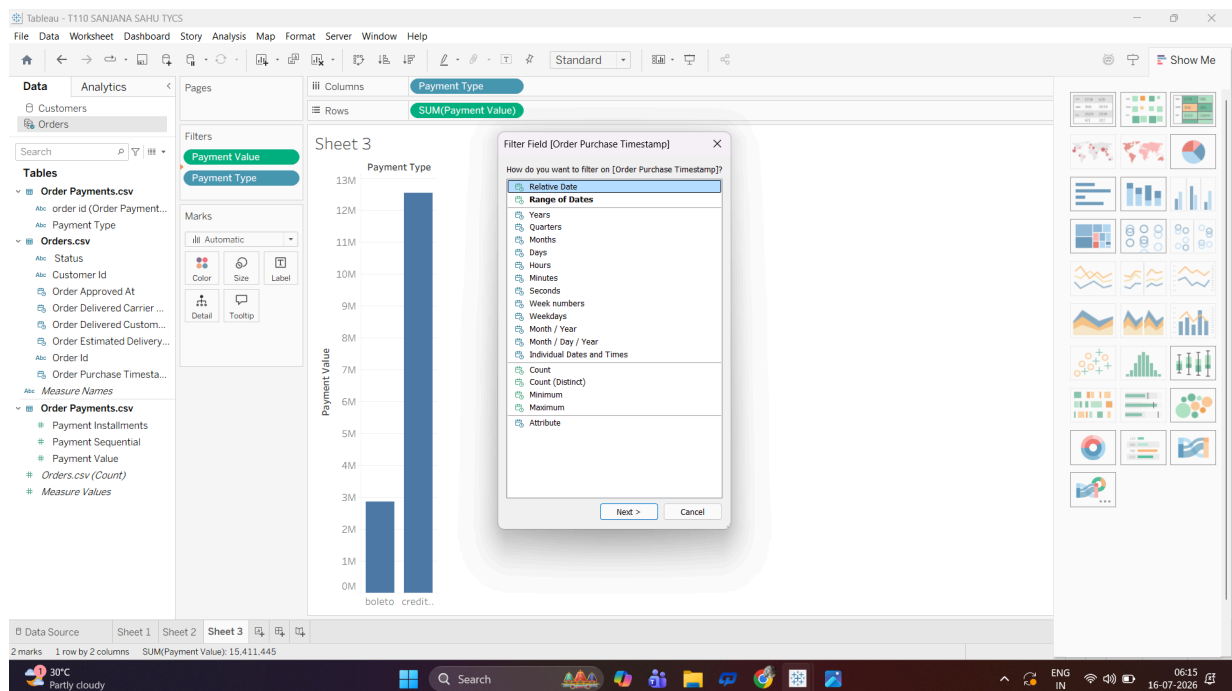
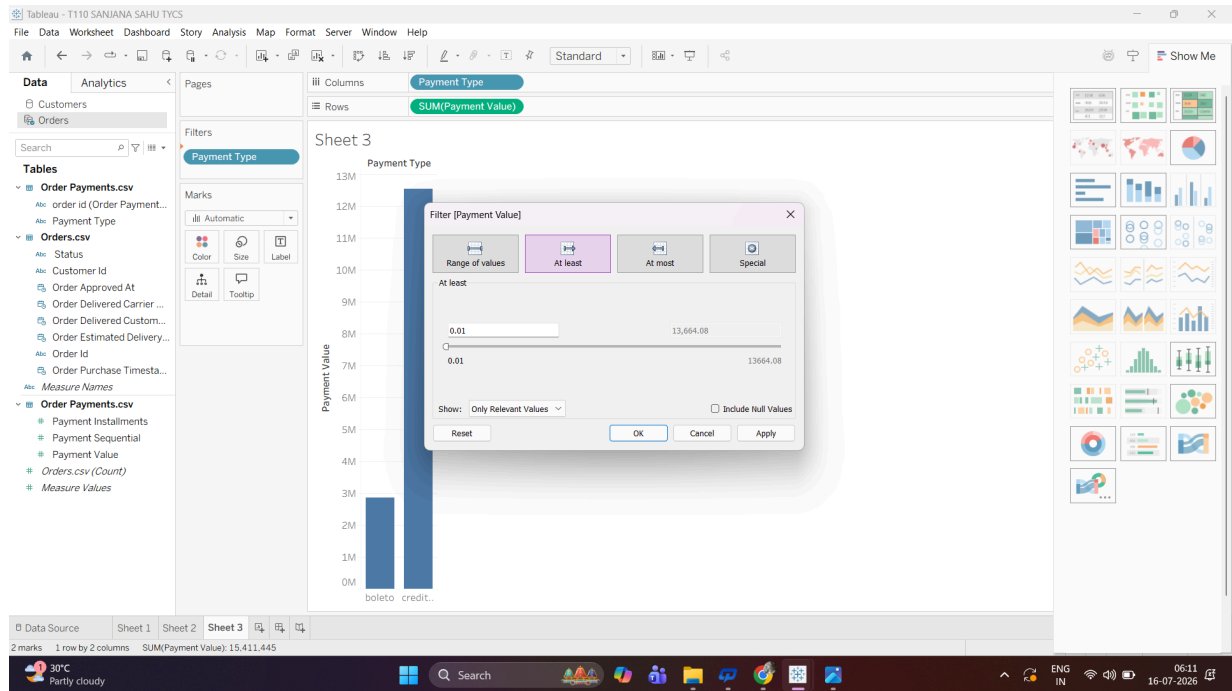
3. Measure Filter: Drag the quantitative numeric variable payment_value out from the measures list and drop it directly onto the Filters shelf. Select All Values in the prompt window, click Next, select the At Least option condition, type in 50 to screen out micro transactions(it will be at 0.1 at by default Change it by clicking on it and typing 50 on it), and click OK.



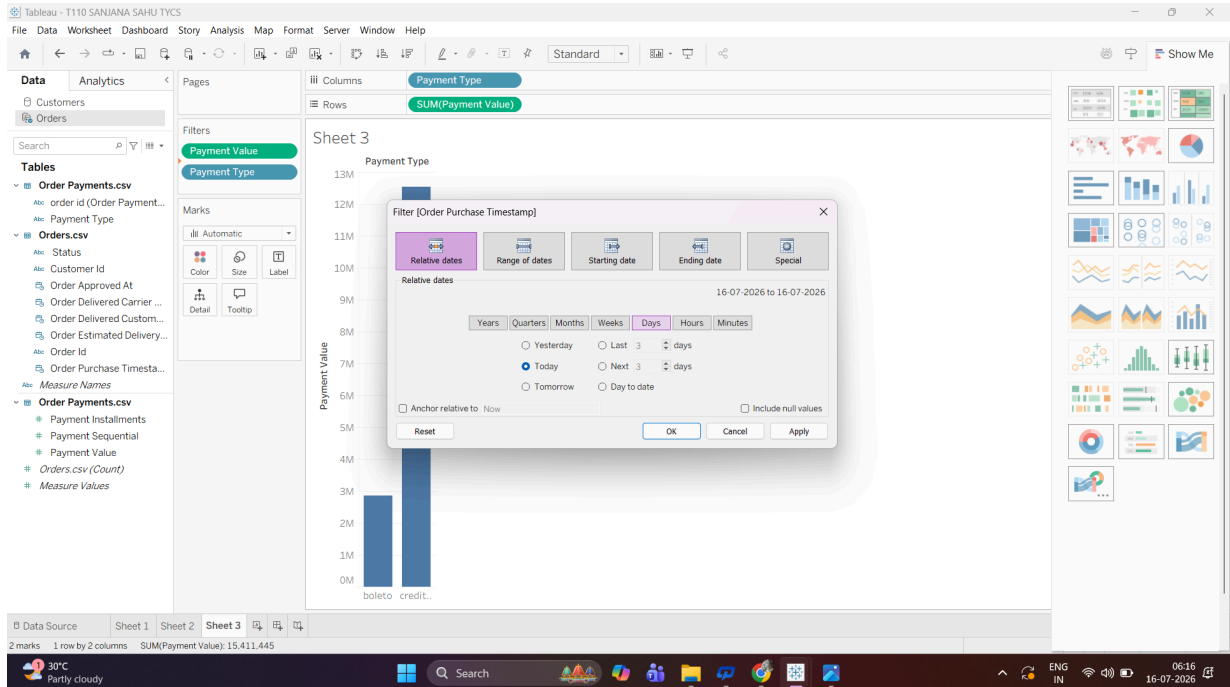
4. Date Filter: Drag the calendar configuration time variable order_purchase_timestamp and drop it onto the Filters shelf. Select Years or Relative Date from the prompt configuration layout page to isolate the visualization into specific historical reporting

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boundaries.



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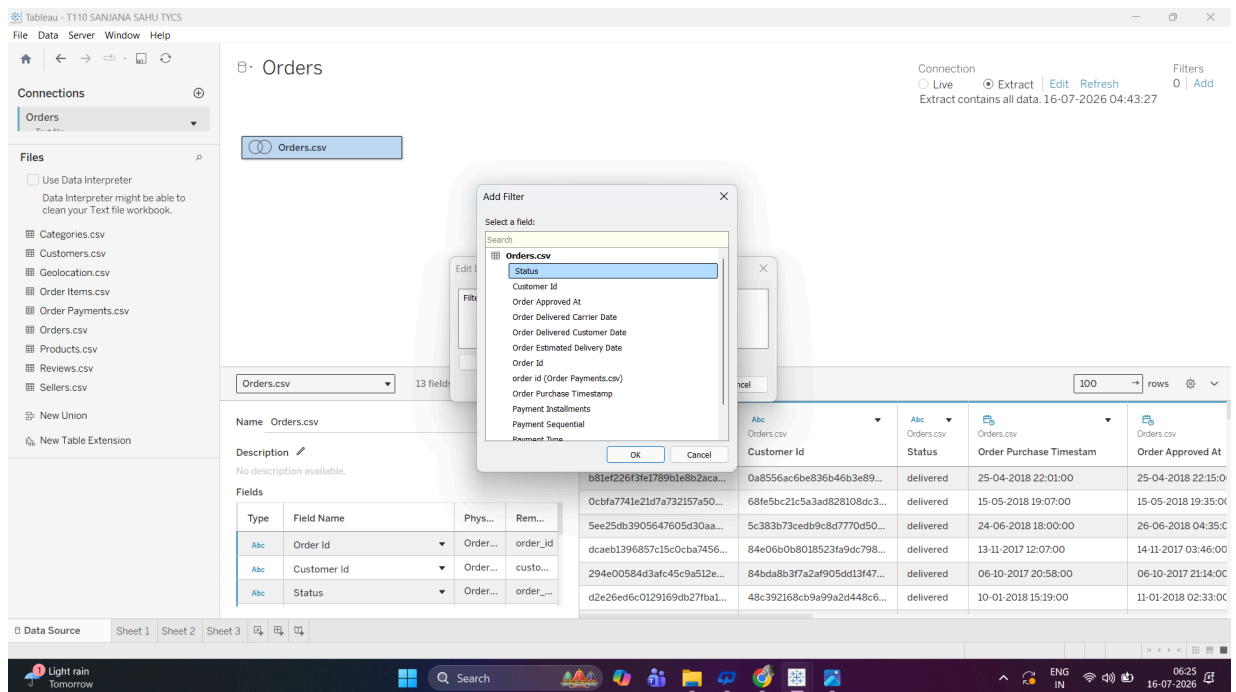
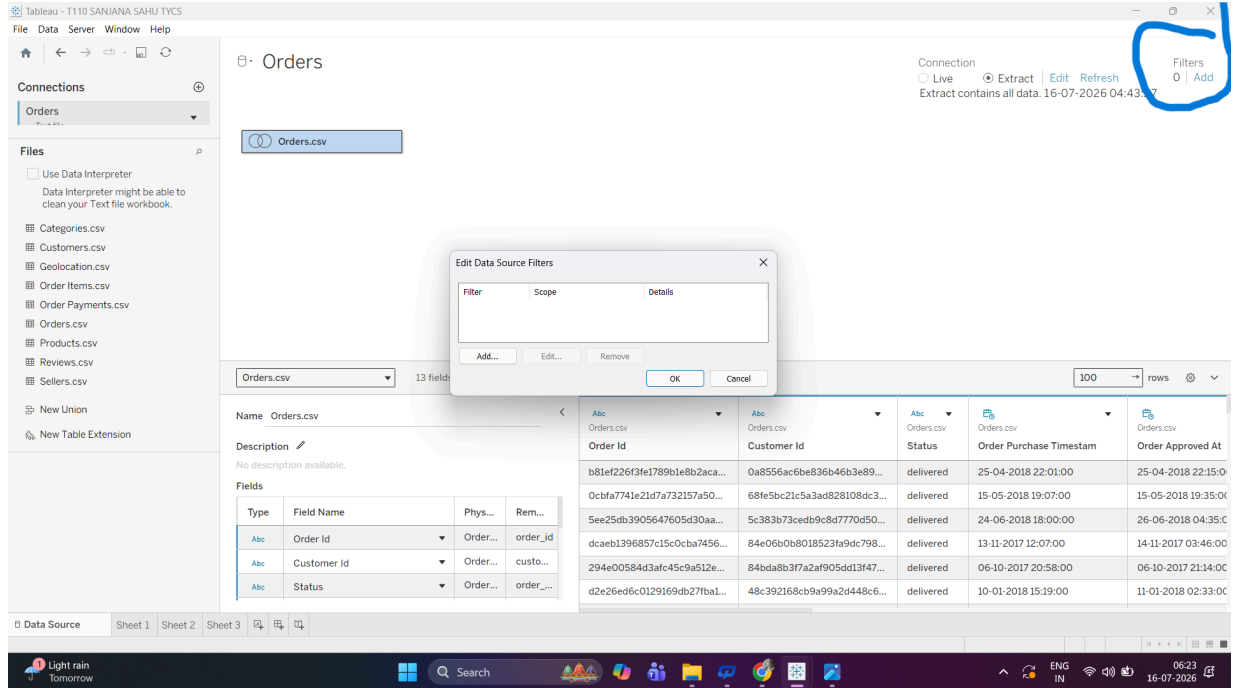
Step G: Optimizing Dashboard Performance

1. Extract Filters: Navigate back to the main Data Source configuration workspace screen. Look immediately below the selected Extract radio dial selection button in the upper-right area and click the small Edit link. Click Add, select the categorical field **Status**, choose to check delivered exclusively, and click OK. Tableau will purge all canceled or unfulfilled order rows from the local extract file entirely.

The screenshot shows the Tableau Data Source configuration workspace for the 'Orders' data source. The 'Extract' radio button is selected, and the 'Edit' link is visible. The 'Fields' section shows a list of fields with their types and descriptions. The 'Status' field is highlighted, and the 'Add' button is visible. The 'Fields' section shows a list of fields with their types and descriptions.

Type	Field Name	Phys...	Rem...
Abc	Order Id	Order...	order_id
Abc	Customer Id	Order...	custo...
Abc	Status	Order...	order_...

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Tableau - T110 SANJANA SAHU TYCS

File Data Server Window Help

Connections

Orders

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Categories.csv

Customers.csv

Geolocation.csv

Order Items.csv

Order Payments.csv

Orders.csv

Products.csv

Reviews.csv

Sellers.csv

New Union

New Table Extension

Orders.csv

13 fields

Name Orders.csv

Description

No description available.

Fields

Type	Field Name
Abc	Order Id
Abc	Customer Id
Abc	Status

Filter [Status]

General Wildcard Condition Top

Select from list Custom value list Use all

Enter search text

- ☐ approved
- ☐ canceled
- ☐ created
- ☒ delivered
- ☐ invoiced
- ☐ processing
- ☐ shipped
- ☐ unavailable

All None Exclude

Summary

Field: [Status]

Selection: Selected 0 of 8 values

Wildcard: All

Condition: None

Limit: None

Reset OK Cancel

100 rows

Order Id	Customer Id	Status	Order Purchase Timestamp	Order Approved At
b81ef226f3fe1789b1e8b2aca...	0a8556ac6be836b46b3e89...	delivered	25-04-2018 22:01:00	25-04-2018 22:15:00
0cbfa7741e21d7a732157a50...	68fe5bc21c5a3ad828108dc3...	delivered	15-05-2018 19:07:00	15-05-2018 19:35:00
See25db3905647605d30aa...	5c383b73cedb9c8d7770d50...	delivered	24-06-2018 18:00:00	26-06-2018 04:35:00
dcaebl396857c15c0c8a7456...	84e06b0b8018523fa9dc798...	delivered	13-11-2017 12:07:00	14-11-2017 03:46:00
294e00584d3afc45c9a512e...	84bda8b3f7a2af905dd13f47...	delivered	06-10-2017 20:58:00	06-10-2017 21:14:00
d2e26ed6c0129169db27ba1...	48c392168cb9a99a2d448c6...	delivered	10-01-2018 15:19:00	11-01-2018 02:33:00

Data Source Sheet 1 Sheet 2 Sheet 3

30°C Partly cloudy

Search

ENG IN

06:26 16-07-2026

Tableau - T110 SANJANA SAHU TYCS

File Data Server Window Help

Connections

Orders

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Use Data Interpreter

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Order Payments.csv

Orders.csv

Products.csv

Reviews.csv

Sellers.csv

New Union

New Table Extension

Orders.csv

13 fields

Name Orders.csv

Description

No description available.

Fields

Type	Field Name	Phys...	Rem...
Abc	Order Id	Order...	order_id
Abc	Customer Id	Order...	custo...
Abc	Status	Order...	order_...

Edit Data Source Filters

Filter Scope Details

Status Orders.csv and related tables keeps 0 of 8 members

Add... Edit... Remove

OK Cancel

100 rows

Order Id	Customer Id	Status	Order Purchase Timestamp	Order Approved At
b81ef226f3fe1789b1e8b2aca...	0a8556ac6be836b46b3e89...	delivered	25-04-2018 22:01:00	25-04-2018 22:15:00
0cbfa7741e21d7a732157a50...	68fe5bc21c5a3ad828108dc3...	delivered	15-05-2018 19:07:00	15-05-2018 19:35:00
See25db3905647605d30aa...	5c383b73cedb9c8d7770d50...	delivered	24-06-2018 18:00:00	26-06-2018 04:35:00
dcaebl396857c15c0c8a7456...	84e06b0b8018523fa9dc798...	delivered	13-11-2017 12:07:00	14-11-2017 03:46:00
294e00584d3afc45c9a512e...	84bda8b3f7a2af905dd13f47...	delivered	06-10-2017 20:58:00	06-10-2017 21:14:00
d2e26ed6c0129169db27ba1...	48c392168cb9a99a2d448c6...	delivered	10-01-2018 15:19:00	11-01-2018 02:33:00

Data Source Sheet 1 Sheet 2 Sheet 3

30°C Partly cloudy

Search

ENG IN

06:26 16-07-2026

The screenshot displays the Tableau Desktop application window titled "Tableau - T110 SANJANA SAHU TYCS". The top menu bar includes File, Data, Server, Window, and Help. On the left sidebar, the "Connections" pane shows "Orders" selected under the "Files" section. Below it, a list of files is visible, including Categories.csv, Customers.csv, Geolocation.csv, Order Items.csv, Order Payments.csv, Orders.csv, Products.csv, Reviews.csv, and Sellers.csv. A checkbox for "Use Data Interpreter" is present. The main workspace area is divided into three sections. The top right corner indicates a "Live" connection to "Orders.csv", which contains 13 fields and 103887 rows of data as of 16-07-2026 04:43:27. The bottom left corner shows the "Data Source" tab with "New Union" and "New Table Extension" options. The central part of the workspace features a preview of the "Orders.csv" data source. It includes a "Description" field stating "No description available." and a "Fields" table. The Fields table lists columns such as Customer Id, Status, Order Purchase Timestamp, and their corresponding physical and remote names. To the right of the Fields table is a data preview grid with five columns: Order Id, Customer Id, Status, Order Purchase Timestamp, and Order Approved At. The grid displays several rows of data, including Order IDs like b81ef226f3fe1789b1e8b2aca... and customer IDs like 68f65bc21c5a3ad828108dc3... The status column shows values like "delivered". The timestamp column shows dates and times like "25-04-2018 22:01:00". The approved at column shows dates and times like "25-04-2018 22:15:00". The bottom of the screen shows the Windows taskbar with various icons and system information.

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